Jaeyoung Yoon, Ph.D.

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EMPLOYMENT

Harvard Medical School 2026 – present
Instructor, Department of Neurology

Boston Children's Hospital 2023 – present
Scientist, F.M. Kirby Neurobiology Center

Massachusetts Institute of Technology 2019 – 2023
Postdoctoral Fellow, McGovern Institute for Brain Research

Seoul National University 2019 – 2019
Postdoctoral Associate, Medical Research Center

EDUCATION

Columbia University 2027 – 2030
J.D. Candidate, Columbia Law School (expected)

Ph.D., Seoul National University 2013 – 2019
School of Biological Sciences, College of Natural Sciences /
Department of Physiology, College of Medicine (*joint affiliation*)
Thesis: *"Short-term synaptic plasticity and persistent activity in the prefrontal cortex"*
(2016 – 2019: Military service, Republic of Korea Army)

B.S., Seoul National University 2009 – 2013
Biological Sciences, College of Natural Sciences

AWARDS AND HONORS

Travel/Research Award, Mind-Brain-Behavior Initiative, Harvard University (2.0 k USD) 2024

Best Presenter Award, F.M. Kirby Neurobiology Center, Boston Children's Hospital (BCH) 2024

Y. Eva Tan Fellowship, Yang-Tan Center for Molecular Therapeutics in Neuroscience, Massachusetts Institute of Technology (MIT) (130.0 k USD) 2021 – 2023

Molecular Therapeutics Impact Report 2020 – 2022, MIT (featured) 2022

BK21 / BK21+ Fellow, National Research Foundation of Korea (NRF) (21.3 k USD) 2013 – 2017

Lecture and Research Scholarship & Merit-based Scholarships, Seoul National University (SNU) 2013 – 2014

Superior Academic Performance Scholarships, SNU 2009 – 2011

SELECTED PUBLICATIONS

Yoon J*, Terauchi A, Castro A, Stone S, Lidov H, Cohen A, Umemori H, Ferguson B*. (2026).
Human cortical and thalamocortical synaptic dysconnectivity in ASD. (*under review*) **(* co-corresponding author)**

Yoon J*, Silva R, Stone S, Lidov H, Osterweil E, Ferguson BR*. (2026).
High-frequency, high-intensity electrical stimulation selectively activates human fast-spiking interneurons.
Frontiers in Synaptic Neuroscience. DOI: 10.3389/fnsyn.2026.1843277 **(* co-corresponding author)**

Cho E, Kwon J, Lee G, Shin J, Lee H, Lee SH, Chung CK*, **Yoon J***, Ho WK*. (2024).
Net synaptic drive of fast-spiking interneurons is inverted towards inhibition in human FCD I epilepsy.
Nature Communications. DOI: 10.1038/s41467-024-51065-7 **(* co-corresponding author)**

Yoon J. (2024). Geometrical determinant of nonlinear synaptic integration in human cortical neurons.
arXiv. DOI: 10.48550/arXiv.2408.05633

Yoon JY, Lee HR, Ho WK, Lee SH. (2020). Disparities in short-term depression among prefrontal cortex synapses sustain persistent activity in a balanced network. *Cerebral Cortex*. DOI: 10.1093/cercor/bhz076

OTHER PUBLICATIONS

Singh A, et al. (2026). FMR1 reduction alters cellular and circuit properties in human cortex. (*under review*)

He M, et al. (2026). Human cortical neuron-based assays to study axon regeneration. (*in preparation*)

Lee BJ, et al. (2025). A specific association of presynaptic K^+ channels with Ca^{2+} channels underlies K^+ channel-mediated regulation of glutamate release. (*under review*)

Yoon J. (2025a). Acute human brain slice preparation for ex vivo electrophysiology and imaging. *protocols.io*. DOI: 10.17504/protocols.io.ewov124qogr2/v1

Yoon J. (2025b). Pulse splitter for 2-photon glutamate uncaging and ex vivo electrophysiology. *protocols.io*. DOI: 10.17504/protocols.io.eq2lyxpjwgx9/v1

Yoon JY, Choi S. (2017). Evidence for presynaptically silent synapses in the immature hippocampus. *Biochemical and Biophysical Research Communications*. DOI: 10.1016/j.bbrc.2016.12.044

INVITED TALKS

"Human cortical and thalamocortical synaptic dysconnectivity in ASD". *Department of Genetics, Harvard Medical School (HMS).* Boston, MA, USA. (Oct 2026)

"Human cortical and thalamocortical synaptic dysconnectivity in ASD". *Department of Convergence IT Engineering, Postech*. Pohang, Korea. (Aug 2026)

Human cortical and thalamocortical synaptic dysconnectivity in ASD". *Gordon Research Seminars (GRS): Fragile X and Autism-Related Disorders*. Newry, ME, USA. (*invited, declined*) (Jun 2026)

"Human cortical and thalamocortical synaptic dysconnectivity in ASD". *Gordon Research Conference (GRC): Thalamocortical Interactions*. Barga (LU), Italy. (Feb 2026)

"Human cortical and thalamocortical synaptic dysconnectivity in ASD". *Department of Neurosurgery, Yonsei University College of Medicine*. Seoul, Korea. (Dec 2025)

"Electrophysiological hallmarks of epilepsy and autism in the human neocortex". *China-Japan-Korea Neuroscience Meeting*. Incheon, Korea. (Aug 2025)

"Thalamocortical synaptic hyperconnectivity in the human neocortex with ASD". *F.M. Kirby Neurobiology Center, BCH*. Boston, MA, USA. (May 2025)

"Net synaptic drive of fast-spiking interneurons is inverted towards inhibition in human FCD I epilepsy". *New England Bioscience Society*. Boston, MA, USA. (Sep 2024)

"Synaptic integration in human neuronal dendrites". *Department of Physiology, SNU College of Medicine*. Seoul, Korea. (Dec 2023)

"Subcellular connectivity and synaptic integration in cortical pyramidal neurons".
Yang-Tan Center for Molecular Therapeutics in Neuroscience & Tan-Yang Center for Autism Research Joint Symposium, McGovern Institute for Brain Research (MIBR), MIT. Cambridge, MA, USA. (Jul 2021)

MEETING ABSTRACTS

Yoon J*, Terauchi A, Castro A, Stone S, Lidov H, Cohen A, Umemori H, Ferguson B* **(* co-corresponding author).** "Human cortical and thalamocortical synaptic dysconnectivity in ASD". *Neuroscience 2026*. Washington, DC, USA. (Nov 2026)

Silva R, **Yoon J**, Ferguson B. "Serotonergic modulation of mouse and human cortical interneurons in acute brain slices". *GRC: Frontal cortex in function, dysfunction, and therapies*. Holderness, NH, USA. (*selected for best poster*) (Aug 2026)

Yoon J*, Terauchi A, Castro A, Stone S, Lidov H, Cohen A, Umemori H, Ferguson B* **(* co-corresponding author).** "Human cortical and thalamocortical synaptic dysconnectivity in ASD". *Federation of European Neuroscience Societies (FENS) Forum*. Barcelona, Spain. (Jul 2026)

Singh A, et al. "Fmr1 reduction drives cell-type transcriptomic remodeling and hyperexcitability in human cortical circuits". *FENS Forum*. Barcelona, Spain. (Jul 2026)

Yoon J*, Terauchi A, Castro A, Cohen A, Umemori H, Ferguson B* **(* co-corresponding author).** "Human cortical and thalamocortical synaptic dysconnectivity in ASD". *GRC: Fragile X and Autism-Related Disorders*. Newry, ME, USA. (Jun 2026)

Yoon J*, Terauchi A, Castro A, Cohen A, Umemori H, Ferguson B* **(* co-corresponding author).** "Human cortical and thalamocortical synaptic dysconnectivity in ASD". *GRC: Thalamocortical Interactions*. Barga (LU), Italy. (Feb 2026)

Yoon J*, Ferguson B* **(* co-corresponding author).** "Functional hyperconnectivity of thalamocortical synapses in human ASD". *Harvard Mind-Brain-Behavior Research Showcase*. Cambridge, MA, USA. (Apr 2025)

Yoon J*, Ferguson B* **(* co-corresponding author).** "Functional hyperconnectivity of thalamocortical synapses in human ASD". *F.M. Kirby Neurobiology Center Retreat, BCH*. Boston, MA, USA. (Mar 2025)

He M, Chung L, Hathaway D, **Yoon J**, Yan T, Yu S, Chalif J, Ferguson B, Osterweil E, He Z. "Human cortical neuron-based assays to study axon regeneration"**.** *F.M. Kirby Neurobiology Center Retreat, BCH*. Boston, MA, USA. (Mar 2025)

Cho E, Kwon J, Lee G, Shin J, Lee H, Lee SH, Chung CK*, **Yoon J***, Ho WK* **(* co-corresponding author).** "Net synaptic drive of fast-spiking interneurons is inverted towards inhibition in human FCD I epilepsy". *GRC: Mechanisms of Epilepsy and Neuronal Synchronization*. Waterville Valley, NH, USA. (Aug 2024)

Yoon JY, Lee HR, Ho WK, Lee SH. "Disparities in short-term depression among prefrontal cortex synapses sustain persistent activity in a balanced network"**.** *Neuro2019*. Niigata, Japan. (Jul 2019)

Yang CH, **Yoon JY**, Ho WK, Lee SH. "Presynaptic mitochondrial calcium release during high-frequency train pulse enhances short-term facilitation"**.** *Korean Society for Brain and Neural Science (KSBNS) Annual Meeting*. Goyang, Korea. (Oct 2016)

TECHNICAL EXPERIENCE

***ex vivo* electrophysiology (patch clamp) in acute human brain slices:** 2019 – present

- Whole-cell (somatic, dendritic, & paired) or excised (outside-out, nucleated, inside-out) patch clamp, in human neocortex and hippocampus
- Human brain slice preparation, from > 100 adult and pediatric patients diagnosed with tumor or epilepsy
- Research Non-Employee Collaborator, Massachusetts General Hospital (MGH) (2021 – 2023)

***ex vivo* electrophysiology (patch clamp) in other applications:** 2013 – present

- Slice electrophysiology setup at BCH/HMS (CLS 13052), MIT MIBR (46-6178), SNU College of Medicine (2-726), and SNU College of Natural Sciences (504-201)
- Patch clamp in rodent brain slices; in neocortex, hippocampus, thalamus, amygdala, and Calyx of Held
- Computational modeling of vesicular dynamics and network biophysics
- Single-cell RNA (scRNA) sequencing with patch clamp (Patch-seq) from human neurons for transcriptomic analysis
- Patch clamp in human brain slice culture, human cortical organoids, and human embryonic kidney cells (HEK 293)

2-photon excitation microscopy (2PEF) and fluorescence microscopy: 2013 – present

- MIT MIBR 2-photon core facility (46-6178) setup and management, including user training (6 postdocs from MIT & Broad Institute of MIT and Harvard trained during 2019 – 2023)
- 2-photon glutamate uncaging (2PGU), with 8x pulse splitter for enhanced 2PEF; including setup and application (provided to University of Ottawa in 2022)
- Intracellular calcium imaging and structural imaging
- Subcellular channelrhodopsin-assisted circuit mapping (sCRACM)

Data analysis and processing: 2013 – present

- Software development for electrophysiology and 2-photon imaging data analysis (https://github.com/flosfor/pvbs)
- Contribution to abfload: MATLAB function for processing .abf file format (https://github.com/fcollman/abfload)

Teaching experience: 2013 – present

- Instructor in Neurology, HMS (2026 – present)
- Mentor (for socially underprivileged students), Postdoctoral Fellow-Student Mentorship Program, BCH (2025 – 2026)
- Teaching Assistant, Data Analysis in Neuroscience Workshop, Interdisciplinary Program in Neuroscience, SNU (2018)
- Teaching Assistant, Biology Lab 1 & 2 / Biological Sciences Research Lab 1 & 2, SNU (2013 – 2014)
- Laboratory personnel training (SNU, MIT, BCH, HMS) (2013 – present)

Other research and laboratory experience: 2013 – present

- *ad hoc* reviewer for *Nature Communications*, *Neuron*, *Cell Reports*, and *Frontiers in Synaptic Neuroscience*
- Technical assistance for *in vivo* patch clamp and Neuropixels setup at MIT MIBR (46-6171)
- Experience with plasmid DNA purification, viral vector packaging, immunohistochemistry, and stereotaxic surgery

LANGUAGES

English (bilingual), Korean (bilingual), Italian (proficient; CILS C2 certified), French (intermediate), German (beginner), MATLAB (proficient)

MEMBERSHIPS

Society for Neuroscience (SfN), Federation of European Neuroscience Societies (FENS), Italian Society for Neuroscience (SINS), Korean Physiological Society, Korean Society for Brain and Neural Sciences (KSBNS), Japan Neuroscience Society, US Chess Federation (USCF; chess.com rating ≤ 2139)